

Yilin Wang (王艺霖)

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EDUCATION

Peking University, School of Physics
Bachelor of Science in Physics; GPA: 3.79/4.0

Sept. 2022 – June 2026 (expected)
Beijing, China

RESEARCH EXPERIENCE

Radiative Effect of Dust on the Climate of Early Mars

Harvard University (Summer Research Intern)

June 2025 – Present

Cambridge, MA, USA

- Advisor: Prof. Robin Wordsworth (Harvard University)
- Calculated key radiative parameters for early Mars dust analogs by applying Mie scattering theory, generating look-up tables for radiative transfer models
- Integrated custom Mie scattering outputs into the PCM_LBL 1D radiative-convective model to investigate the climatic effect of airborne dust on early Mars
- Modeled the reflectance spectra of regolith using reflectance spectroscopy theory in order to constrain surface albedo, which determines both the radiative energy balance and the observational characteristics

Stability of Polar Vortices in Planetary Atmospheres

Peking University

June 2024 – Sept. 2025

Beijing, China

- Advisors: Prof. Tao Cai (Macau Univ. of Science and Tech.), Prof. Jun Yang (Peking University)
- Investigated the stability of planetary polar vortices through 3D simulations and linear stability analysis to identify critical conditions leading to vortex splitting
- Explored the relationship between vortex instability and key atmospheric parameters (Ro, Fe), identified the gravito-inertial instability mechanism

Cloud-Resolving Simulation of Precipitation in Planetary Atmospheres

Peking University

Mar. 2025 – June 2025

Beijing, China

- Advisor: Prof. Jun Yang (Peking University)
- Utilized the System for Atmospheric Modeling (SAM) to conduct cloud-resolving simulations of diverse planetary atmospheres
- Analyzed the influence of varying solar spectra and planet surface gravity on precipitation patterns and atmospheric dynamics

Planetary Atmospheric Spectrum Retrieval based on Machine Learning

Peking University

Nov. 2024 – Dec. 2024

Beijing, China

- Independent course project for Numerical Simulation and AI Forecast of Geophysics Fluids
- Engineered a comprehensive synthetic dataset of planetary emission spectra for model training and validation using the pyratbay radiative transfer model
- Designed, trained, and benchmarked six distinct machine learning architectures (including Linear Regression, Random Forest, MLP, CNNs, and Transformer) to comparatively analyze their performance in retrieving atmospheric composition and profiles

PUBLICATION

1. **Wang, Y.**, Cai, T., & Yang, J., Constraints on the depth of gas giants' polar vortices from gravito-inertial instability ([Manuscript in preparation, draft available](#)).

PRESENTATIONS

2024 National Conference in Planetary Science

Stability of Polar Vortices in Planetary Atmospheres

Oct. 2024

Nanjing, Jiangsu, China

- Outstanding Student Presentation Award

2024 Lin-Bridge Exoplanet Symposium

Stability of Polar Vortices in Planetary Atmospheres

Aug. 2024

Zhuhai, Guangdong, China

- Third Prize

TEACHING EXPERIENCE

Fundamentals of Planetary Science

School of Earth and Space Sciences, Peking University

Fall 2024

Beijing, China

- Teaching Assistant | Graded weekly assignments and led problem-solving sessions

AWARDS & HONORS

- 2023 - May 4th Scholarship, Peking University (The highest university-level honor for students)
- 2023 - Merit Student Pacesetter, Peking University (Top-tier award for academic excellence, awarded to approx. top 2% of students)
- 2023 - National Second Prize, China Undergraduate Physics Tournament (CUPT)
- 2021 - Gold Medal (Top 70 nationwide), 38th Chinese Physics Olympiad (CPhO) Finals

SELECTED COURSES

- **Core Physics:** Theoretical Mechanics, Electrodynamics, Equilibrium Statistical Physics, Quantum Mechanics
- **Mathematical & Computational:** Methods of Mathematical Physics, Statistical Methods in Experimental Physics, Computational Physics
- **Planetary & Atmospheric Science:** Planetary Physics, Fluid Mechanics, Fundamentals in Atmospheric Physics, Numerical Simulation and AI Forecast of Geophysical Fluids, EPS 244 (Harvard Univ.) Atmospheric Evolution and Habitability of Terrestrial Planets

SKILLS

- **Programming Languages:** Python, Fortran, C, MATLAB, NCL, Mathematica
- **Data Analysis / ML:** NumPy, SciPy, Matplotlib, scikit-learn, PyTorch
- **Climate Models:** SAM, LMDZ Generic PCM, PCM_LBL
- **Utilities:** Linux, $\text{\LaTeX}$
- **Languages:** Mandarin (Native), English (Proficient, TOEFL 111)